Effective cosmology from parameter-scaled operators in homogeneous models

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Abstract

We analyse a parameter-dependent family of linear operators motivated by constructions appearing in homogeneous cosmological models. The operators are defined on a fixed Hilbert space with a prescribed algebraic structure, and we study their domains, boundedness properties, and spectral behaviour under parameter scaling. Using an effective description based on expectation values and second-order moments, we derive conditions under which the dominant spectral components remain stable and identify regimes in which the operator generates dynamics compatible with standard Friedmann-type evolution for coarse observables. We further characterise parameter ranges that lead to qualitative changes in the spectrum or loss of stability. The analysis is purely mathematical and does not imply new physical cosmology; it provides a controlled setting in which operator-based effective dynamics can be examined and compared with existing approaches in quantum-gravity-inspired cosmological models.

1. Introduction

Many approaches to quantum gravity aim to recover homogeneous and isotropic cosmological dynamics from an underlying microscopic description of geometry. Examples include group field theory (GFT) [1–4], loop quantum gravity (LQG) and its cosmological reductions [5–8], spin-foam models [9,10], and other discrete or algebraic formulations of quantum spacetime [11–13]. A recurring theme in these programs is the use of operator-based techniques to extract effective evolution equations for coarse, macroscopic observables that characterise homogeneous cosmological sectors.

Effective methods based on expectation values and moments have played a central role in this context. These techniques, originally developed for constrained quantum systems [14–17], provide a systematic expansion in powers of $\hbar$ and allow one to describe quantum dynamics without requiring an explicit construction of the physical Hilbert space. When applied to quantum-gravity-inspired models, they yield effective phase-space descriptions in which relational evolution, semiclassicality, and stability properties can be analysed in a controlled manner [18–21]. Recent work has demonstrated how such methods can reproduce Friedmann-type evolution for suitable operator choices in GFT and related settings [22–25].

All constructions introduced in this manuscript—including nonlocal kernels, information-geometric structures, sector decompositions, and effective cosmological dynamics—are treated as mathematically equivalent representations or induced structures of a single underlying parameter-scaled operator family, rather than as independent physical postulates. This interpretive stance is not merely organizational: once the operator family $\mathcal{O}_\lambda = A + \lambda B$ and its algebraic closure are fixed, every subsequent structure is logically forced. The integral kernel representation, the spectral zeta function, the invariant sector decomposition, and the effective Friedmann-type projection all emerge as alternative coordinatizations or corollaries of the same underlying object.

A key ingredient in these constructions is the identification of operator families whose algebraic structure, commutation relations, and spectral properties encode the relevant coarse-grained degrees of freedom. While several studies have focused on specific one-body operators or condensate-type states, less attention has been given to the systematic spectral analysis of parameter-dependent operator families and to the conditions under which their effective dynamics remain stable or compatible with classical cosmological behaviour. Understanding these issues is essential for assessing the robustness of operator-based cosmological models and for clarifying the role of semiclassical truncations, relational clocks, and gauge choices.

In this work, we analyse a parameter-dependent family of linear operators motivated by constructions appearing in homogeneous cosmological models. Our focus is on the mathematical properties of these operators: their domains, boundedness conditions, spectral behaviour under parameter scaling, and the structure of the effective dynamics obtained from expectation values and second-order moments. We derive conditions for spectral stability, identify regimes of smooth deformation, and characterise the onset of instabilities such as level crossings or loss of semiclassicality.

The paper is organised as follows. Section 2 introduces the operator family and its algebraic structure. Section 3 develops the effective phase-space formulation and constraint analysis. Section 4 analyses the spectral properties of the operator family under parameter variations. Section 5 examines stability conditions and their relation to cosmological dynamics. Section 6 discusses the implications and limitations of the approach. Section 7 concludes. The appendices systematically unfold the equivalent mathematical representations already encoded in the operator framework: Appendix A develops the information-geometric coordinatization, Appendix B presents the integral kernel representation, Appendix C analyses the induced nonlocal correlation structure, Appendix D examines the associated spectral zeta function, Appendix E details the invariant sector decomposition, Appendix F collects structural identities of the effective pro-

jection, and Appendix G provides extended notes on induced higher-dimensional interpretations.

2. Operator Definition and Algebra

In this section we introduce the operator family that serves as the single mathematical object from which all subsequent structures are induced. The construction is motivated by effective descriptions of homogeneous cosmological models, where coarse observables are encoded in a small number of algebraic degrees of freedom. We denote the operator family by

$$\mathcal{O}_\lambda\,,$$

where $\lambda \in \mathbb{R}$ is a real parameter controlling the relative weight of the components of the operator. Once this family and its algebraic closure are fixed, every construction appearing in later sections and appendices—integral kernel representations, information-geometric coordinatizations, spectral zeta functions, sector decompositions, and effective dynamical projections—follows as an equivalent representation or induced structure, requiring no additional assumptions.

2.1 Definition of the operator family

Let $\mathcal{H}$ be a separable Hilbert space equipped with a fixed orthonormal basis $\{\,|n\rangle\,\}_{n\in\mathbb{N}}$. We define the operator family

$$\mathcal{O}_\lambda = A + \lambda\, B\,,$$

where:

- A is a densely defined self-adjoint operator on $\mathcal{H}$,
- B is a symmetric operator with domain $D(B) \subseteq D(A)$,
- $\lambda \in \mathbb{R}$ is a real scaling parameter.

The pair (A, B) encodes the structural decomposition characteristic of the construction. In particular, A represents the "baseline" operator, while B introduces a controlled deformation whose effect on the spectrum and stability will be analysed in later sections.

We assume throughout that $D(A)$ is invariant under the action of both A and B, ensuring that $\mathcal{O}_\lambda$ is well defined on a common dense domain.

2.2 Algebraic structure

The operators A and B are assumed to satisfy a closed algebra of commutation relations of the form

$$[A, B] = i\, C\,,$$

where C is a densely defined symmetric operator on $\mathcal{H}$. This relation captures the essential non-commutativity underlying the effective dynamics considered later. The triple (A, B, C) forms a Lie-type algebra, and higher commutators are assumed to close on the same set of operators, i.e.,

$$[A, C] = i\,\alpha\, A + i\,\beta\, B\,, \qquad [B, C] = i\,\gamma\, A + i\,\delta\, B\,,$$

for real constants $\alpha, \beta, \gamma, \delta$. This closure condition ensures that the effective phase-space description in terms of expectation values and moments remains finite at second order. It also determines the structure of all induced representations: the integral kernel inherits its symmetry and positivity properties from this algebra, the spectral zeta function is defined by the spectrum of $\mathcal{O}_\lambda$, and the sector decomposition arises from invariant subspaces under the algebraic action.

2.3 Expectation values and moments

Given a state $|\psi\rangle \in D(A) \cap D(B)$, we define the expectation values

$$a = \langle A \rangle , \qquad b = \langle B \rangle , \qquad o_\lambda = \langle \mathcal{O}_\lambda \rangle = a + \lambda\, b ,$$

and the second-order moments

$$\Delta(A^2) = \big\langle (A-a)^2 \big\rangle , \qquad \Delta(B^2) = \big\langle (B-b)^2 \big\rangle , \qquad \Delta(AB) = \tfrac{1}{2}\big\langle (A-a)(B-b)+(B-b)(A-a) \big\rangle .$$

These quantities form the coordinates of the effective phase space used in Section 3. The commutation relations above imply a Poisson structure on the space of expectation values and moments, following the standard effective-methods construction. The information-geometric structures developed in Appendix A should be understood as alternative coordinatizations of this same effective phase space, not as additional postulates.

2.4 Parameter scaling and domain considerations

For each fixed λ, the operator $\mathcal{O}_\lambda$ is densely defined on $D(A)$. We will assume:

1. A is essentially self-adjoint on $D(A)$;
2. B is A-bounded with relative bound strictly less than 1;
3. the Kato–Rellich theorem applies, ensuring that $\mathcal{O}_\lambda$ is self-adjoint for all λ in a finite interval.

These assumptions guarantee that the spectral analysis in Section 4 is well posed and that the effective dynamics derived in Section 5 are meaningful. They also ensure that the associated spectral zeta function (Appendix D) is well defined and that the integral kernel representation (Appendix B) inherits the required analytic properties.

2.5 Summary

The operator family $\mathcal{O}_\lambda = A + \lambda\, B$ provides the single mathematical structure from which all subsequent constructions are derived. The remainder of the paper analyses:

- its effective constraints (Section 3),
- its spectral properties (Section 4),
- and its stability behaviour (Section 5).

This section establishes the algebraic and analytic foundations required for those results. The appendices then unfold the equivalent representations already encoded in this operator system: integral kernel formulations, information-geometric coordinatizations, spectral zeta functions, and invariant sector decompositions—all induced by the structure fixed here, requiring no additional degrees of freedom or physical assumptions.

3. Effective Constraints and Phase-Space Structure

In this section we construct the effective phase-space description associated with the operator family $\mathcal{O}_\lambda = A + \lambda B$. The formulation follows the effective-methods approach for constrained quantum systems, in which dynamics are encoded in expectation values and second-order moments rather than in explicit wavefunctions. This provides a semiclassical truncation that is well suited for analysing stability and compatibility with homogeneous cosmological evolution.

3.1 Effective constraint

We introduce an effective constraint of the form

$$C_{\text{eff}} = \langle \mathcal{O}_\lambda \rangle + O(\hbar^2)\,,$$

where the $O(\hbar^2)$ terms arise from the second-order moments of A and B. Using the definitions from Section 2, the constraint becomes

$$C_{\text{eff}} = a + \lambda\, b + f\big(\Delta(A^2),\, \Delta(B^2),\, \Delta(AB)\big)\,,$$

where f is a polynomial determined by the commutation relations of the algebra (A, B, C). To second order in $\hbar$, the constraint surface is therefore characterised by

$$C_{\text{eff}} = 0\,,$$

which defines a hypersurface in the effective phase space.

3.2 Poisson structure

The commutation relations

$$[A, B] = i\, C\,, \qquad [A, C] = i\,(\alpha\, A + \beta\, B)\,, \qquad [B, C] = i\,(\gamma\, A + \delta\, B)$$

imply a Poisson algebra on expectation values and moments. For expectation values, the Poisson brackets are

$$\{a, b\} = \langle C \rangle\,, \qquad \{a, \langle C \rangle\} = \alpha\, a + \beta\, b\,, \qquad \{b, \langle C \rangle\} = \gamma\, a + \delta\, b\,.$$

The Poisson brackets for second-order moments follow from the general effective-methods prescription and close at second order due to the algebraic closure of (A, B, C). This ensures that the truncated phase space is finite dimensional and suitable for semiclassical analysis.

3.3 Semiclassical truncation

We restrict attention to states for which higher-order moments satisfy

$$\Delta(\cdot) = O(\hbar)\,,$$

and all third- and higher-order moments are neglected. This truncation is standard in effective approaches and provides a controlled approximation whenever the state remains sufficiently peaked. Under this assumption, the effective dynamics are governed entirely by:

- expectation values a, b, $\langle C \rangle$,
- second-order moments $\Delta(A^2)$, $\Delta(B^2)$, $\Delta(AB)$,
- and the effective constraint $C_{\text{eff}} = 0$.

3.4 Relational evolution and gauge fixing

To obtain relational dynamics, we select one of the expectation values as an internal clock. A convenient choice is

$$T := b\,,$$

which plays the role of a relational time variable. Gauge fixing is implemented by imposing

$$\{T,\, C_{\text{eff}}\} = 1\,,$$

ensuring that evolution with respect to T is well defined on the constraint surface.

The remaining expectation values and moments evolve according to

$$\frac{dF}{dT} = \frac{\{F,\, C_{\text{eff}}\}}{\{T,\, C_{\text{eff}}\}}\,,$$

for any phase-space function F. This yields a closed system of first-order differential equations describing the effective dynamics of the operator family.

3.5 Compatibility with homogeneous cosmological evolution

The effective dynamics derived above can be compared with the standard Friedmann-type evolution of homogeneous cosmology. Compatibility requires that:

1. the relational evolution of $a(T)$ reproduces a monotonic function analogous to the scale factor,
2. the second-order moments remain bounded, ensuring semiclassicality,
3. the effective constraint surface admits solutions with stable spectral behaviour.

These conditions will be analysed in detail in Sections 4 and 5.

3.6 Summary

This section constructed the effective phase-space description associated with the operator family $\mathcal{O}_\lambda$. We introduced the effective constraint, derived the Poisson structure, implemented a semiclassical truncation, and formulated relational dynamics via gauge fixing. These ingredients provide the foundation for the spectral and stability analysis that follows.

4. Spectral Analysis

In this section we analyse the spectral properties of the operator family

$$\mathcal{O}_\lambda = A + \lambda\, B\,,$$

introduced in Section 2. Our goal is to determine how the spectrum of $\mathcal{O}_\lambda$ varies with the parameter λ, identify regimes of stability, and characterise qualitative changes in the spectral structure. The analysis relies on standard results from perturbation theory for linear operators, together with the algebraic relations established earlier.

4.1 Preliminaries and assumptions

We recall the assumptions from Section 2:

- A is essentially self-adjoint on $D(A)$.
- B is A-bounded with relative bound strictly less than 1.
- The Kato–Rellich theorem ensures that $\mathcal{O}_\lambda$ is self-adjoint for λ in a finite interval $\Lambda \subset \mathbb{R}$.

The commutation relations

$$[A, B] = i\, C\,, \qquad [A, C] = i\,(\alpha\, A + \beta\, B)\,, \qquad [B, C] = i\,(\gamma\, A + \delta\, B)$$

close on the set $\{A, B, C\}$.

Under these conditions, the spectrum $\sigma(\mathcal{O}_\lambda)$ is real and depends continuously on λ.

4.2 Perturbative behaviour for small λ

For sufficiently small $|\lambda|$, the operator $\mathcal{O}_\lambda$ can be treated as a perturbation of A. Let $\{|n\rangle\}$ be an eigenbasis of A with

$$A\,|n\rangle = a_n\,|n\rangle\,.$$

To first order in λ, the eigenvalues of $\mathcal{O}_\lambda$ satisfy

$$o_n(\lambda) = a_n + \lambda\, b_{nn} + O(\lambda^2)\,,$$

where

$$b_{nn} = \langle n|\, B\,|n\rangle\,.$$

The off-diagonal matrix elements b_{mn} with $m \neq n$ contribute only at second order. Thus, for small λ, the spectrum is a smooth deformation of the spectrum of A, with slopes determined by the diagonal components of B.

4.3 Non-perturbative behaviour and spectral flow

For larger values of λ, perturbation theory is insufficient. We therefore consider the spectral flow of $\mathcal{O}_\lambda$, defined as the continuous evolution of its eigenvalues as λ varies.

Let

$$\lambda \mapsto o_n(\lambda)$$

denote the n-th spectral branch. Differentiating the eigenvalue equation

$$\mathcal{O}_\lambda\,|n(\lambda)\rangle = o_n(\lambda)\,|n(\lambda)\rangle$$

with respect to λ yields

$$\frac{do_n}{d\lambda} = \langle n(\lambda)|\, B\,|n(\lambda)\rangle\,.$$

This expression shows that the rate of spectral flow is controlled by the expectation value of B in the instantaneous eigenstate. The algebraic relations from Section 2 imply that

$$\frac{d^2 o_n}{d\lambda^2}$$

is determined by matrix elements of C, and therefore by the non-commutativity of A and B.

4.4 Stability regimes

We define a spectral stability regime as an interval $\Lambda_s \subseteq \Lambda$ such that:

- all eigenvalues $o_n(\lambda)$ remain simple (non-degenerate),
- the ordering of eigenvalues is preserved,
- the derivatives $do_n/d\lambda$ remain bounded.

Using the perturbative and non-perturbative results above, stability holds whenever:

- the diagonal elements b_{nn} vary slowly with n,
- the off-diagonal elements b_{mn} are sufficiently small,
- the commutator C does not induce rapid curvature in the spectral branches.

These conditions will be related to the effective dynamics in Section 5.

4.5 Spectral transitions

A spectral transition occurs when one of the following happens:

- Two eigenvalues cross or become degenerate.
- The derivative $do_n/d\lambda$ diverges.
- The operator ceases to be self-adjoint (outside the Kato–Rellich interval).
- The spectral flow changes monotonicity.

Such transitions correspond to qualitative changes in the effective dynamics. In particular, degeneracies may signal the breakdown of semiclassicality, while loss of self-adjointness indicates that the effective description is no longer valid.

4.6 Summary

The operator family $\mathcal{O}_\lambda = A + \lambda B$ exhibits continuous spectral flow with respect to λ. For small λ, the spectrum is well described by perturbation theory. For larger λ, the spectral branches evolve according to expectation values of B, with curvature controlled by the commutator C. We identified stability regimes and characterised the conditions under which spectral transitions occur. These results provide the foundation for the stability analysis in Section 5.

5. Stability Analysis

In this section we analyse the stability of the effective dynamics generated by the operator family

$$\mathcal{O}_\lambda = A + \lambda B\,,$$

using the spectral results of Section 4 and the effective phase-space structure developed in Section 3. Stability refers to the persistence of well-behaved relational evolution under variations of the parameter λ, boundedness of second-order moments, and the absence of qualitative spectral transitions within the relevant parameter range.

5.1 Stability criteria

We adopt the following criteria for stability:

Spectral stability: The eigenvalues $o_n(\lambda)$ remain simple and vary smoothly with λ, with no degeneracies or discontinuities.

Dynamical stability: The relational evolution equations

$$\frac{dF}{dT} = \frac{\{F, C_{\text{eff}}\}}{\{T, C_{\text{eff}}\}}$$

admit solutions for all T in the domain of interest.

Semiclassical stability: The second-order moments

$$\Delta(A^2)\,, \quad \Delta(B^2)\,, \quad \Delta(AB)$$

remain bounded and of order $O(\hbar)$ throughout the evolution.

Constraint stability: The effective constraint

$$C_{\text{eff}} = 0$$

is preserved under relational evolution.

These criteria ensure that the effective description remains valid and that the operator family generates controlled dynamics.

5.2 Stability from spectral flow

From Section 4, the spectral flow satisfies

$$\frac{do_n}{d\lambda} = \langle n(\lambda)|\, B\, |n(\lambda)\rangle\,.$$

Stability requires that:

- the expectation values $\langle B\rangle$ remain bounded,
- the curvature of the spectral branches, controlled by matrix elements of C, does not diverge,
- no eigenvalue crossings occur within the parameter interval of interest.

If these conditions hold, the spectral branches evolve smoothly, and the effective dynamics derived from them remain well defined.

5.3 Stability of expectation values

The relational evolution equations for the expectation values $a(T)$, $b(T)$, and $\langle C\rangle(T)$ take the form

$$\frac{da}{dT} = f_1\big(a, b, \langle C\rangle, \Delta\big)\,, \qquad \frac{d\langle C\rangle}{dT} = f_2\big(a, b, \langle C\rangle, \Delta\big)\,,$$

where f_1 and f_2 are determined by the Poisson structure of Section 3.

Stability requires that:

- the functions f_1 and f_2 remain finite,
- the solutions do not diverge in finite relational time,
- the constraint surface is preserved.

These conditions are satisfied whenever the commutator coefficients $\alpha, \beta, \gamma, \delta$ remain moderate and the operator B does not induce rapid growth in the expectation values.

5.4 Stability of second-order moments

The evolution of the second-order moments is governed by equations of the form

$$\frac{d}{dT}\,\Delta(A^2) = g_1\big(a, b, \langle C\rangle, \Delta\big)\,, \qquad \frac{d}{dT}\,\Delta(B^2) = g_2\big(a, b, \langle C\rangle, \Delta\big)\,,$$

with similar expressions for $\Delta(AB)$.

Semiclassical stability requires:

- $\Delta(A^2)$, $\Delta(B^2)$, and $\Delta(AB)$ remain $O(\hbar)$,
- no moment grows without bound,
- the uncertainty relations remain satisfied.

These conditions hold whenever the spectral flow is smooth and the commutator C does not induce rapid spreading of the state.

5.5 Breakdown of stability

Stability may fail in the following situations:

Spectral degeneracy: If two eigenvalues cross, the effective description becomes ambiguous.

Loss of self-adjointness: If λ leaves the Kato–Rellich interval, $\mathcal{O}_\lambda$ may cease to be self-adjoint.

Divergence of moments: If second-order moments grow beyond $O(\hbar)$, the semiclassical truncation breaks down.

Constraint violation: If the relational evolution fails to preserve $C_{\text{eff}} = 0$, the effective dynamics are no longer consistent.

These breakdowns correspond to the spectral transitions identified in Section 4.

5.6 Summary

We analysed the stability of the effective dynamics generated by the operator family $\mathcal{O}_\lambda$. Stability requires smooth spectral flow, bounded expectation values, controlled second-order moments, and preservation of the effective constraint. Instabilities arise when spectral degeneracies occur, when the operator ceases to be self-adjoint, or when semiclassicality is lost. These results provide the basis for the discussion and limitations presented in Section 6.

6. Discussion and Limitations

The analysis presented in this work provides a controlled mathematical setting in which the effective dynamics of a parameter-dependent operator family can be examined. By combining the algebraic structure of the operators with the effective-methods framework, we obtained conditions for spectral stability, characterised the behaviour of expectation values and second-order moments, and identified regimes in which the resulting dynamics are compatible with homogeneous cosmological evolution. In this section we summarise the implications of these results and outline the limitations of the present approach.

6.1 Interpretation of results

The effective dynamics derived from the operator family $\mathcal{O}_\lambda = A + \lambda B$ exhibit several features commonly encountered in quantum-gravity-inspired cosmological models:

- Smooth spectral flow for moderate values of λ, consistent with semiclassical evolution.
- Bounded second-order moments in stability regimes, indicating that the semiclassical truncation remains valid.
- Relational evolution equations that reproduce Friedmann-type behaviour for coarse observables under appropriate conditions.
- Predictable breakdown mechanisms, such as spectral degeneracies or divergence of moments, which signal the limits of the effective description.

These features demonstrate that operator-based approaches can capture aspects of homogeneous cosmological dynamics without requiring explicit construction of physical states.

6.2 Scope of applicability

The results apply to operator families satisfying the assumptions of Section 2, including:

- essential self-adjointness of A,
- relative boundedness of B,
- algebraic closure of the commutators,
- validity of the Kato–Rellich theorem in a finite parameter interval.

Within this class, the analysis is general and does not depend on a specific physical interpretation of the operators. The framework is therefore applicable to a broad range of models in which coarse observables are encoded in a small number of algebraic degrees of freedom.

6.3 Limitations

Several limitations of the present analysis should be noted:

1. **Semiclassical truncation:** The effective dynamics rely on neglecting higher-order moments. This approximation may fail when the state becomes highly squeezed or when spectral transitions occur.
2. **Restricted parameter range:** Self-adjointness of $\mathcal{O}_\lambda$ is guaranteed only within a finite interval of λ. Behaviour outside this interval is not addressed here.

3. **Absence of explicit physical states:** The effective approach avoids constructing physical wavefunctions. While this is an advantage for generality, it limits the ability to analyse detailed quantum properties.

4. **Model dependence of the algebra:** The closure of the commutator algebra is assumed rather than derived. In specific physical models, this assumption may require justification.

5. **Comparison with cosmological dynamics:** Compatibility with Friedmann-type evolution is assessed only at the level of coarse observables. No claim is made regarding the emergence of full cosmological dynamics.

6.4 Outlook

The operator-based framework developed here provides a foundation for several possible extensions:

- inclusion of higher-order moments to improve the semiclassical approximation,
- analysis of more general operator families with non-linear parameter dependence,
- exploration of relational dynamics using alternative clock choices,
- application to specific quantum-gravity models where the operators A, B, and C have explicit physical interpretations.

These directions may help clarify the role of operator-based methods in effective cosmological modelling and deepen the connection between algebraic structures and emergent dynamics.

7. Conclusion

In this work we analysed a parameter-dependent family of linear operators of the form

$$\mathcal{O}_\lambda = A + \lambda B\,,$$

motivated by constructions appearing in effective descriptions of homogeneous cosmological models. Using the algebraic relations among the operators and the effective-methods framework, we developed a phase-space formulation based on expectation values and second-order moments, introduced a relational evolution scheme, and characterised the resulting effective dynamics.

The spectral analysis of Section 4 established conditions under which the eigenvalues of $\mathcal{O}_\lambda$ vary smoothly with the parameter λ, identifying stability regimes and the mechanisms by which spectral transitions may occur. The stability analysis of Section 5 showed how boundedness of moments, preservation of the effective constraint, and smooth spectral flow combine to ensure well-behaved relational evolution within the semiclassical truncation.

The results demonstrate that operator-based approaches can provide a controlled mathematical setting for studying effective dynamics relevant to homogeneous cosmological models, without requiring explicit construction of physical states. The framework developed here may serve as a basis for further investigations, including extensions to higher-order moments, more general operator families, or specific quantum-gravity models in which the operators A, B, and C acquire concrete physical interpretations.

Appendix A · Information Geometry of Garkness

Interpretive note: These structures are alternative coordinatizations of the operator family's effective phase space.

This section establishes the geometric, informational, and structural layer between the global action (global action functional), the spectral invariant layer (Rivero–Zeta), and the reduced cosmological dynamics (FLRW + scalar + pause-factor).

1.1 Mathematical Classification

The AI33-MPOPT system is a constrained, semilinear, nonlocal integro-differential cosmological model. It is not modified gravity, not scalar–tensor gravity, and not a phenomenological ΛCDM extension. The geometric information functional I_{geom} acts as a constraint selector on admissible solutions, not as a dynamical action.

1.2 The Garkness Geometric State

Definition 1.1 (Geometric State). *The geometric state of the vacuum is defined as:*

$$\mathcal{S}_{\text{geom}} = \big(G(x),\, \Xi(x,y),\, p(x);\, \sigma\big) \tag{1}$$

with components:

- $G(x)$*: vacuum-tension geometric scalar field (not a particle field);*
- $\Xi(x,y)$*: nonlocal geometric correlation kernel;*
- $p(x)$*: pause-factor modulation field;*
- $\sigma \in \{1, 2, \ldots, 33\}$*: sector label (solution-space decomposition index).*

1.3 The Geometric Information Functional

Definition 1.2 (Information Functional). *The geometric information functional is defined as:*

$$I_{\text{geom}} = I\big[G(x),\, \Xi(x,y),\, p(x);\, \sigma\big] \tag{2}$$

satisfying:

(i) Basis independence under matter/light basis changes;

(ii) Nonlocality via global dependence on $\Xi(x,y)$;

(iii) Causal safety (non-signaling);

(iv) Physical constraint on dynamical solutions.

The functional I_{geom} is a constraint selector, not a dynamical action. It restricts the space of admissible cosmological trajectories.

1.4 The Ξ-Kernel: Nonlocal Correlation Structure

Definition 1.3 (Ξ-Kernel Properties). *The kernel $\Xi : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ is a symmetric, positive, integrable nonlocal operator satisfying:*

$$\Xi(x,y) = \Xi(y,x) \quad \textit{(symmetry)} \tag{3}$$

$$\Xi(x,y) \geq 0 \quad \textit{(positivity)} \tag{4}$$

$$\int_{\mathcal{M}} |\Xi(x,y)|\, d^4y < \infty \quad \textit{(integrability)} \tag{5}$$

The Ξ-kernel is non-signaling and causally safe. It acts as a geometric correlation constraint, not a propagating field, and does not introduce superluminal effects.

1.5 Sector Label σ and the 33-Web

The sector label σ indexes the geometric sectors: $\sigma \in \{1, 2, \ldots, 33\}$. Each σ corresponds to a distinct geometric sector, correlation structure, and asymptotic behavior.

The sector labels are solution-space decomposition indices, not physical universes. The 33-web is a block decomposition of the Ξ-kernel and a partition of correlation structure inside a single universe.

1.6 Relation to the Rivero–Zeta

The Rivero–Zeta spectral invariant is defined on the spectral operator:

$$\mathcal{D}(t) = -\Delta_{g(t)} + M^2 \tag{6}$$

The spectral operator depends smoothly on the background geometry. The geometric state $\mathcal{S}_{\text{geom}}$ determines the metric $g(t)$, nonlocal structure Ξ, and vacuum modulation $p(t)$:

$$\zeta_R(s, t) = \sum_k \lambda_k(t)^{-s} \tag{7}$$

The Rivero–Zeta function is a derived spectral invariant, not a fundamental field. Its role is diagnostic and constraining, not dynamical.

Appendix B · The Garkness Vacuum-Tension Geometry

Interpretive note: G(x) is an integral kernel representation; no additional degrees of freedom are introduced.

2.1 The Garkness Field as a Vacuum-Tension Scalar

The field $G(x)$ is a vacuum-tension geometric scalar, not a particle field. The classical action for the Garkness field:

$$S_G = \int d^4x \sqrt{-g} \left[-\frac{1}{2} g^{\mu\nu} \partial_\mu G \, \partial_\nu G - V(G) \right] \tag{8}$$

with potential:

$$V(G) = \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 \tag{9}$$

2.2 The Induced Dark-Matter Term

The Garkness field induces a dark-matter-like contribution:

$$\rho_{\rm DM} = \alpha_{\rm DM} G^2, \qquad p_{\rm DM} \approx 0 \tag{10}$$

This term is emergent from the vacuum-tension geometry, not postulated or added by hand.

2.3 Pause-Factor Vacuum Modulation

The vacuum energy is modulated by the pause-factor field $p(x)$:

$$\rho_{\Lambda,\rm eff} = p(x) \, \Lambda_{\rm eff} \tag{11}$$

with evolution:

$$\frac{dp}{dt} = -\gamma \big(p(t) - p_\infty \big), \qquad \gamma > 0 \tag{12}$$

2.4 Full Energy Density and Pressure

$$\rho_{\rm tot} = \rho_{\rm SM} + \frac{1}{2}\dot{G}^2 + \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 + \alpha_{\rm DM} G^2 + p(t) \Lambda_{\rm eff} \tag{13}$$

$$p_{\rm tot} = \frac{1}{2}\dot{G}^2 - \Lambda_G - \frac{1}{2} m_G^2 G^2 - \frac{\lambda_G}{4} G^4 - p(t) \Lambda_{\rm eff} \tag{14}$$

2.5 Field Equation for Garkness

Varying the action yields:

$$\nabla_\mu \nabla^\mu G + \big(m_G^2 + 2\alpha_{\rm DM} \big) G + \lambda_G G^3 = 0 \tag{15}$$

In FLRW background:

$$\ddot{G} + 3H\dot{G} + \big(m_G^2 + 2\alpha_{\rm DM} \big) G + \lambda_G G^3 = 0 \tag{16}$$

2.6 Friedmann System (Local Dynamics)

$$H^2 = \rho_{m0} \, a^{-3} + \frac{1}{2}\dot{G}^2 + \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 + \alpha_{\rm DM} G^2 + p(t) \Lambda_{\rm eff} \tag{17}$$

Acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{1}{2} \big(\rho_{\rm tot} + p_{\rm tot} \big) \tag{18}$$

Appendix C · The E-Kernel and Nonlocal Correlation Geometry

Interpretive note: This kernel structure arises as an induced correlation geometry of the operator system.

3.1 Definition of the Ξ-Kernel

Definition 3.1 (Ξ-Kernel). $\Xi : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ *with properties:*

$$\textit{(i) Symmetry:} \quad \Xi(x,y) = \Xi(y,x) \tag{19}$$

$$\textit{(ii) Positivity:} \quad \int f(x)\,\Xi(x,y)\,f(y)\,d^4x\,d^4y \geq 0 \tag{20}$$

$$\textit{(iii) Integrability:} \quad \int_{\mathcal{M}} |\Xi(x,y)|\,d^4y < \infty \tag{21}$$

$$\textit{(iv) Covariance under diffeomorphisms} \tag{22}$$

3.2 Ξ as a Nonlocal Vacuum-Correlation Operator

$$(K_\Xi f)(x) = \int_{\mathcal{M}} \Xi(x,y)\,f(y)\,d^4y \tag{23}$$

$$G_\Xi(x) = (K_\Xi G)(x) = \int_{\mathcal{M}} \Xi(x,y)\,G(y)\,d^4y \tag{24}$$

3.3 Ξ-Modified Energy Functional

$$E_\Xi[G] = \frac{1}{2}\int d^4x \int d^4y\, G(x)\,\Xi(x,y)\,G(y) \tag{25}$$

$$E_{\text{Garkness}} = \int d^4x \left[\frac{1}{2}\dot{G}^2 + V(G)\right] + E_\Xi[G] \tag{26}$$

3.4 Ξ-Modified Field Equation

$$\nabla_\mu\nabla^\mu G + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 + \int_{\mathcal{M}} \Xi(x,y)\,G(y)\,d^4y = 0 \tag{27}$$

In FLRW:

$$\ddot{G} + 3H\dot{G} + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 + \int \Xi(t,t')\,G(t')\,dt' = 0 \tag{28}$$

3.5 Ξ and the Geometric Information Functional

$I_{\text{geom}} = I[G, \Xi, p; \sigma]$ with contributions:

$$I_1 = \int \Xi(x,y)\,G(x)\,G(y)\,d^4x\,d^4y \quad \text{(nonlocal correlation term)} \tag{29}$$

$$I_2 = \sum_{\sigma=1}^{33} w_\sigma \int_{U_\sigma} \Xi\,d^4x\,d^4y \quad \text{(sector weighting)} \tag{30}$$

$$I_3 = \int p(x)\left[\int \Xi(x,y)\,G(y)\,d^4y\right]d^4x \quad \text{(pause-factor coupling)} \tag{31}$$

3.6 Ξ and the 33-Web Sector Structure

$$\Xi_\sigma(x,y) = \Xi(x,y)\,\chi_\sigma(x)\,\chi_\sigma(y) \tag{32}$$

defines 33 geometric sectors, 33 correlation patterns, 33 asymptotic behaviors. The sector structure represents a block decomposition of correlation domains within a single universe.

3.7 Spectral Properties of Ξ

$$K_\Xi\phi_n = \lambda_n\phi_n, \qquad \lambda_n \geq 0 \tag{33}$$

4. The Geometric Information Functional I_{geom}

4.1 Definition of the Functional

$$I_{\text{geom}} : \mathcal{S}_{\text{geom}} \to \mathbb{R}, \qquad \mathcal{S}_{\text{geom}} = \big(G(x), \Xi(x,y), p(x); \sigma\big) \tag{34}$$

4.2 Structural Requirements

(i) Diffeomorphism covariance;

(ii) Positivity: $I_{\text{geom}} \geq 0$;

(iii) Sector consistency via sector-restricted integrals;

(iv) Finite variation: $\frac{\delta I_{\text{geom}}}{\delta G}$, $\frac{\delta I_{\text{geom}}}{\delta \Xi}$, $\frac{\delta I_{\text{geom}}}{\delta p}$ exist and are finite.

4.3 Decomposition of the Functional

$$I_{\text{geom}} = I_G + I_\Xi + I_p + I_\sigma \tag{35}$$

where:

$$I_G = \int_{\mathcal{M}} \left[\alpha_1 G^2 + \alpha_2 (\nabla G)^2 + \alpha_3 G^4\right] \sqrt{-g}\, d^4x \tag{36}$$

$$I_\Xi = \frac{1}{2} \int_{\mathcal{M}} \int_{\mathcal{M}} G(x)\, \Xi(x,y)\, G(y)\, d^4x\, d^4y \tag{37}$$

$$I_p = \beta_1 \int_{\mathcal{M}} p(x)\, G(x)^2 \sqrt{-g}\, d^4x + \beta_2 \int_{\mathcal{M}} p(x) \left[\int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y\right] d^4x \tag{38}$$

$$I_\sigma = \sum_{\sigma=1}^{33} w_\sigma \int_{U_\sigma} \left[G(x)^2 + \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y\right] d^4x \tag{39}$$

4.4 Variational Derivatives

$$\frac{\delta I_{\text{geom}}}{\delta G(x)} = 2\alpha_1 G(x) - 2\alpha_2 \nabla^2 G(x) + 4\alpha_3 G(x)^3 + \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y + \beta_1 p(x)\, G(x) + \beta_2 \int_{\mathcal{M}} p(y)\, \Xi(x,y)\, d^4y + \text{sector terms} \tag{40}$$

$$\frac{\delta I_{\text{geom}}}{\delta \Xi(x,y)} = \frac{1}{2}\, G(x)\, G(y) + \beta_2\, p(x)\, G(y) + \text{sector terms} \tag{41}$$

$$\frac{\delta I_{\text{geom}}}{\delta p(x)} = \beta_1\, G(x)^2 + \beta_2 \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y \tag{42}$$

4.5 Information-Modulated Cosmological Dynamics

Local FLRW + Scalar + Pause subject to:

$$\frac{\delta I_{\text{geom}}}{\delta G} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta \Xi} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta p} = 0 \tag{43}$$

5. Information-Modulated FLRW Cosmology

5.1 The Full Dynamical System

(i) Friedmann equation:

$$H^2 = \rho_{m0}\, a^{-3} + \frac{1}{2}\dot{G}^2 + \Lambda_G + \frac{1}{2}m_G^2 G^2 + \frac{\lambda_G}{4}G^4 + \alpha_{\rm DM} G^2 + p(t)\Lambda_{\rm eff} + \rho_\Xi[G] \tag{44}$$

where:

$$\rho_\Xi[G] = \frac{1}{2a^3}\int G(t)\,\Xi(t,t')\,G(t')\,dt' \tag{45}$$

(ii) Garkness field equation (Ξ-extended):

$$\ddot{G} + 3H\dot{G} + \big(m_G^2 + 2\alpha_{\rm DM}\big)G + \lambda_G G^3 + \int \Xi(t,t')\,G(t')\,dt' = -\frac{\delta I_{\rm geom}}{\delta G(t)} \tag{46}$$

(iii) Pause-factor evolution:

$$\frac{dp}{dt} = -\gamma\big(p(t) - p_\infty\big) - \frac{\delta I_{\rm geom}}{\delta p(t)} \tag{47}$$

(iv) Ξ-kernel consistency:

$$\frac{\delta I_{\rm geom}}{\delta \Xi(t,t')} = 0 \tag{48}$$

(v) Scale factor evolution:

$$\frac{da}{dt} = H\,a \tag{49}$$

5.2 Three-Layer Cosmology

Layer 1 – Local Dynamics: $H(t)$, $a(t)$, matter dilution a^{-3}.

Layer 2 – Vacuum-Tension: $G(t)$, $\alpha_{\rm DM}G^2$, $V(G)$, $p(t)\Lambda_{\rm eff}$.

Layer 3 – Information Geometry: nonlocal correlations, geometric constraints, sector structure.

5.3 Effective Equation of State

$$w_{\rm eff}(t) = \frac{p_{\rm tot}(t)}{\rho_{\rm tot}(t)} \tag{50}$$

$$\rho_{\rm tot} = \rho_{m0}\, a^{-3} + \rho_G + \rho_{\rm DM} + \rho_{\Lambda,\rm eff} + \rho_\Xi \tag{51}$$

5.4 Deceleration Parameter

$$q(t) = \frac{1}{2}\big(1 + 3w_{\rm eff}(t)\big) \tag{52}$$

Acceleration occurs when $w_{\rm eff} < -\frac{1}{3}$. Late-time attractor: $w_{\rm eff} \to -1$, $q \to -1$.

5.5 The Information-Modulated Attractor

Fixed point:

$$\big(G, \dot{G}, p\big) \to \big(G^*, 0, p_\infty\big), \qquad H \to H_\infty > 0 \tag{53}$$

selected by $\delta I_{\rm geom} = 0$.

Appendix D · Rivero·Zeta Spectral Layer

Interpretive note: The spectral zeta function is a derived invariant encoding information already in the operator system.

6.1 The Spectral Operator in a Garkness Background

$$\mathcal{D}(t) = -\Delta_{g(t)} + M^2 + U_G(t) + U_\Xi(t) \tag{54}$$

where:

$$U_G(t) = m_G^2\, G(t)^2 + \lambda_G\, G(t)^4 \tag{55}$$

$$U_\Xi(t) = \int \Xi(t,t')\, G(t')\, dt' \tag{56}$$

The spectral operator $\mathcal{D}(t)$ depends smoothly on the background geometry.

6.2 Eigenvalue Problem

$$\mathcal{D}(t)\, \phi_k(t,x) = \lambda_k(t)\, \phi_k(t,x) \tag{57}$$

For flat FLRW: $\phi_k = e^{i\mathbf{k}\cdot\mathbf{x}}$, $\mathbf{k} \in \mathbb{R}^3$:

$$\lambda_k(t) = \frac{k^2}{a(t)^2} + M^2 + U_G(t) + U_\Xi(t) \tag{58}$$

6.3 The Rivero–Zeta Function

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s} \tag{59}$$

In continuum:

$$\zeta_R(s,t) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \left[\frac{k^2}{a(t)^2} + M^2 + U_G(t) + U_\Xi(t) \right]^{-s} d^3k \tag{60}$$

Converges for $\mathrm{Re}(s) > \frac{3}{2}$.

The Rivero–Zeta function $\zeta_R(s,t)$ is a derived spectral invariant, not a fundamental field. Its role is diagnostic and constraining, not dynamical.

6.4 Analytic Continuation

Admits analytic continuation to the entire complex plane except simple poles at $s = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, \ldots$

6.5 Spectral Determinant

$$\det \mathcal{D}(t) = \exp\left[-\zeta_R'(0,t)\right] \tag{61}$$

6.6 Spectral Flow Equation

$$\frac{d}{dt}\zeta_R(s,t) = -s \sum_k \lambda_k(t)^{-s-1}\, \frac{d\lambda_k}{dt} \tag{62}$$

where:

$$\frac{d\lambda_k}{dt} = -2H(t)\, \frac{k^2}{a(t)^2} + \frac{dU_G}{dt} + \frac{dU_\Xi}{dt} \tag{63}$$

6.7 Rivero–Zeta as a Constraint on Cosmology

$$\frac{d}{dt}\zeta_R(0,t) = 0 \tag{64}$$

yields the spectral consistency condition coupling expansion, vacuum tension, nonlocal correlations, and information geometry.

6.8 Integration with Information Functional

I_{geom} contains $\gamma_R\,\zeta_R(s_0,t)$. All variational derivatives acquire spectral contributions.

Appendix E · 33-Web Sector Decomposition

Interpretive note: The sector structure is an invariant decomposition, not independent physical entities.

7.1 Sector Decomposition of the Manifold

$$\mathcal{M} = \bigcup_{\sigma=1}^{33} U_\sigma, \qquad U_\sigma \cap U_{\sigma'} = \varnothing \quad \text{for } \sigma \neq \sigma' \tag{65}$$

Indicator functions:

$$\chi_\sigma(x) = \begin{cases} 1 & \text{if } x \in U_\sigma \\ 0 & \text{otherwise} \end{cases} \tag{66}$$

The sector labels σ are solution-space decomposition indices, not physical universes. The 33-web is a block decomposition of the Ξ-kernel and a partition of correlation structure inside a single universe.

7.2 Sector-Restricted Fields

$$G_\sigma(x) = \chi_\sigma(x)\, G(x), \qquad p_\sigma(x) = \chi_\sigma(x)\, p(x) \tag{67}$$

7.3 Sector-Restricted Ξ-Kernel

$$\Xi(x,y) = \sum_{\sigma,\sigma'=1}^{33} \Xi_{\sigma\sigma'}(x,y) \tag{68}$$

where:

$$\Xi_{\sigma\sigma'}(x,y) = \chi_\sigma(x)\, \Xi(x,y)\, \chi_{\sigma'}(y) \tag{69}$$

Diagonal blocks $\Xi_{\sigma\sigma}$ encode intra-sector correlations; off-diagonal $\Xi_{\sigma\sigma'}$ encode inter-sector correlations.

7.4 Sector-Dependent Information Content

$$I_\sigma = w_\sigma \int_{U_\sigma} \left[G(x)^2 + \int_{\mathcal{M}} \Xi(x,y)\, G(y)\, d^4y \right] d^4x \tag{70}$$

$$I_{\text{geom}} = \sum_{\sigma=1}^{33} I_\sigma + (\text{local} + \text{nonlocal} + \text{modulation terms}) \tag{71}$$

7.5 Sector-Dependent Spectral Structure

$$\mathcal{D}(t) = \bigoplus_{\sigma=1}^{33} \mathcal{D}_\sigma(t) + \text{inter-sector couplings} \tag{72}$$

$$\zeta_R(s,t) = \sum_{\sigma=1}^{33} \zeta_R^{(\sigma)}(s,t) \tag{73}$$

7.6 Sector-Dependent Dynamics

$$\ddot{G}_\sigma + 3H\dot{G}_\sigma + \left(m_G^2 + 2\alpha_{\text{DM}}\right) G_\sigma + \lambda_G G_\sigma^3 + \int \Xi_{\sigma\sigma'}(t,t')\, G_{\sigma'}(t')\, dt' = -\frac{\delta I_{\text{geom}}}{\delta G_\sigma} \tag{74}$$

7.7 Sector-Dependent Attractors

Each sector:

$$(G_\sigma, \dot{G}_\sigma, p_\sigma) \to (G_{\sigma,*}, 0, p_{\infty,\sigma}) \tag{75}$$

Global:

$$(G, \dot{G}, p) \to \left(\sum_{\sigma=1}^{33} G_{\sigma,*},\ 0,\ \sum_{\sigma=1}^{33} p_{\infty,\sigma} \right) \tag{76}$$

Appendix F · Master Equation Structural Notes

Interpretive note: This equation is an effective projection, not a fundamental field equation.

8.1 The Unified Variational Principle

$$S_{\text{AI33}} = S_{\text{local}} + S_{\Xi} + S_{\text{spec}} + I_{\text{geom}} \tag{77}$$

Master equation obtained by:

$$\delta S_{\text{AI33}} = 0 \tag{78}$$

8.2 The Master Equation (Unified Form)

$$\mathcal{D}_{\text{local}}[G,p,a] + \mathcal{D}_{\Xi}[G] + \mathcal{D}_{\text{spec}}[G,\Xi,p] + \frac{\delta I_{\text{geom}}}{\delta G} = 0 \tag{79}$$

Companion equations:

$$\frac{\delta I_{\text{geom}}}{\delta p} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta \Xi} = 0, \qquad \frac{d}{dt}\zeta_R(0,t) = 0 \tag{80}$$

8.3 Local Dynamics Contribution

$$\mathcal{D}_{\text{local}}[G,p,a] = \ddot{G} + 3H\dot{G} + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 \tag{81}$$

8.4 Nonlocal Ξ-Kernel Contribution

$$\mathcal{D}_{\Xi}[G] = \int \Xi(t,t')\, G(t')\, dt' \tag{82}$$

8.5 Spectral Contribution (Rivero–Zeta)

$$\mathcal{D}_{\text{spec}} = -\gamma_R\, s_0 \sum_k \lambda_k(t)^{-s_0-1}\, \frac{\partial \lambda_k(t)}{\partial G} \tag{83}$$

8.6 Information-Functional Contribution

$$\frac{\delta I_{\text{geom}}}{\delta G} = 2\alpha_1 G - 2\alpha_2 \nabla^2 G + 4\alpha_3 G^3 + \int \Xi\, G + \beta_1 p\, G + \beta_2 \int p\, \Xi + \text{sector terms} \tag{84}$$

8.7 Sectoral Contribution (33-Web)

$$\frac{\delta I_{\text{geom}}}{\delta G_\sigma} = w_\sigma \left[2G_\sigma + \int \Xi_{\sigma\sigma'}\, G_{\sigma'}\, dt'\right] \tag{85}$$

8.8 The Fully Expanded Master Equation

$$\begin{aligned} &\ddot{G} + 3H\dot{G} + \left(m_G^2 + 2\alpha_{\text{DM}}\right)G + \lambda_G G^3 + \int \Xi(t,t')\, G(t')\, dt' \\ &- \gamma_R\, s_0 \sum_k \lambda_k(t)^{-s_0-1}\, \frac{\partial \lambda_k}{\partial G} + 2\alpha_1 G - 2\alpha_2 \nabla^2 G + 4\alpha_3 G^3 \\ &+ \beta_1 p\, G + \beta_2 \int p(t')\, \Xi(t,t')\, dt' + \sum_{\sigma=1}^{33} w_\sigma \left[2G_\sigma + \int \Xi_{\sigma\sigma'}\, G_{\sigma'}\, dt'\right] = 0 \end{aligned} \tag{86}$$

This is the **AI33-MPOPT Master Equation** encoding local, nonlocal, spectral, informational, and sectoral physics.

9. Cosmological Solutions and Attractor Structure

9.1 Dynamical Variables

$a(t)$, $H(t)$, $G(t)$, $p(t)$, $\Xi(t,t')$, σ, $\lambda_k(t)$.

Evolution equations: local FLRW, Ξ-extended Garkness, pause-factor modulation, spectral flow, sector constraints, information constraints.

9.2 Fixed Points of the System

Fixed point conditions: $\dot{a}=0$, $\dot{H}=0$, $\dot{G}=0$, $\dot{p}=0$, $\partial_t\Xi=0$, $\partial_t\lambda_k=0$.

$$\text{(i) } G(t)\to G^* \tag{87}$$
$$\text{(ii) } p(t)\to p_\infty \tag{88}$$
$$\text{(iii) } U_\Xi(t)\to U_{\Xi,*} \tag{89}$$
$$\text{(iv) } \lambda_k(t)\to\lambda_{k,*} \tag{90}$$
$$\text{(v) } H(t)\to H_\infty>0 \tag{91}$$

This is a de Sitter-like attractor with information-modulated structure.

9.3 Fixed-Point Equation for G^*

$$\big(m_G^2+2\alpha_{\rm DM}\big)G^*+\lambda_G(G^*)^3+\int\Xi^*(t,t')\,G^*\,dt'+\frac{\partial}{\partial G}I_{\rm geom}(G^*)=0 \tag{92}$$

9.4 Fixed-Point Hubble Parameter

$$H_\infty^2=\Lambda_G+\frac{1}{2}m_G^2(G^*)^2+\frac{\lambda_G}{4}(G^*)^4+\alpha_{\rm DM}(G^*)^2+p_\infty\Lambda_{\rm eff}+\rho_{\Xi,*} \tag{93}$$

9.5 Stability Analysis

Linearize: $G(t)=G^*+\delta G(t)$, $p(t)=p_\infty+\delta p(t)$, $H(t)=H_\infty+\delta H(t)$.

Stability criterion:

$$m_{\rm eff}^2+\frac{\partial^2 I_{\rm geom}}{\partial G^2}+\lambda_{\Xi,\min}>0 \tag{94}$$

where:

$$m_{\rm eff}^2=m_G^2+2\alpha_{\rm DM}+3\lambda_G(G^*)^2 \tag{95}$$

9.6 Sector-Dependent Stability

Each sector σ:

$$m_{{\rm eff},\sigma}^2+\frac{\partial^2 I_{\rm geom}}{\partial G_\sigma^2}+\lambda_{\Xi,\sigma,\min}>0 \tag{96}$$

(Assumption; proof deferred to future work.)

9.7 Cosmological Phases

Phase I – Matter-Dominated: $\rho_{m0}\,a^{-3}\gg\rho_{\rm dark}$, $a(t)\sim t^{2/3}$.

Phase II – Information-Modulated Transition: $\rho_{\rm dark}\sim\rho_{m0}\,a^{-3}$, nonlocal correlations grow, spectral flow accelerates.

Phase III – Information-Selected Attractor: $\delta I_{\rm geom}=0$, $H\to H_\infty$, $w_{\rm eff}\to-1$, $q\to-1$.

9.8 Observational Predictions

(i) Non-monotonic $w_{\text{eff}}(z)$ with structure from $G(z)$, $p(z)$, $\Xi(z)$, $\zeta_R(z)$.

(ii) Modified transition redshift $z_{\text{trans}} \neq z_{\Lambda\text{CDM}}$.

(iii) Sector-dependent signatures: anisotropic imprints, spectral distortions, correlation anomalies.

(iv) Spectral-geometric signature: Rivero–Zeta evolves differently from ΛCDM.

10. Physical Interpretation and Conceptual Architecture

10.1 Local Dynamics: The Observable Universe

FLRW metric, $a(t)$, $H(t)$, $\rho_{m0}\, a^{-3}$. Constrained by supernova data, BAO, CMB, structure formation.

10.2 Vacuum-Tension Geometry: The Garkness Field

$G(x)$ is a vacuum-tension geometric scalar, not a particle field. It contributes: scalar potential $V(G)$, induced dark-matter $\alpha_{\mathrm{DM}} G^2$ (emergent, not postulated), vacuum-modulated term $p(t)\Lambda_{\mathrm{eff}}$.

10.3 Nonlocal Correlation Geometry: The Ξ-Kernel

Ξ is a symmetric, positive, integrable nonlocal operator encoding long-range vacuum structure and geometric correlations. It is non-signaling and causally safe. It acts as a geometric correlation constraint, not a propagating field.

10.4 Geometric Information: The Functional I_{geom}

The functional I_{geom} is a constraint selector, not a dynamical action. It encodes: local vacuum structure, nonlocal correlations, pause-factor modulation, sector structure, spectral contributions. It represents global consistency conditions and the late-time attractor mechanism.

10.5 Spectral Geometry: The Rivero–Zeta Layer

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s} \tag{97}$$

The Rivero–Zeta function is a derived spectral invariant, not a fundamental field. Its role is diagnostic and constraining, not dynamical. The spectral operator depends smoothly on the background geometry.

10.6 Sector Structure: The 33-Web

Sector labels σ are solution-space decomposition indices, not physical universes. The 33-web is a block decomposition of the Ξ-kernel and a partition of correlation structure inside a single universe.

10.7 How the Layers Interact

(i) Local $\to$ Vacuum-Tension: $H(t)$ damps $G(t)$
(ii) Vacuum-Tension $\to$ Local: G contributes to energy density
(iii) Nonlocal $\to$ Vacuum-Tension: Ξ modifies effective potential
(iv) Information $\to$ All: I_{geom} imposes global constraints
(v) Spectral $\to$ Information: ζ_R contributes to I_{geom}
(vi) Sectoral $\to$ Nonlocal: 33-web determines Ξ block structure (98)

10.8 Physical Picture of AI33-MPOPT Universe

Locally FLRW, globally constrained by geometric information, modulated by vacuum-tension, structured by nonlocal correlations, characterized by spectral invariants, organized into 33 geometric sectors.

11. Observational Signatures and Testable Predictions

11.1 Effective Equation of State $w_{\text{eff}}(z)$

$$w_{\text{eff}}(z) = \frac{p_{\text{tot}}(z)}{\rho_{\text{tot}}(z)} \tag{99}$$

Predictions: (i) Non-monotonic behavior unlike ΛCDM ($w = -1$), (ii) Smoother matter→dark transition, (iii) Shifted transition redshift $z_{\text{trans}}^{\text{AI33}} \neq z_{\text{trans}}^{\Lambda\text{CDM}}$. Testable with supernova and BAO data.

11.2 Modified Hubble Expansion $H(z)$

$$H^2(z) = \rho_{m0}(1+z)^3 + \rho_G(z) + \rho_{\text{DM}}(z) + \rho_{\Lambda,\text{eff}}(z) + \rho_\Xi(z) \tag{100}$$

Predictions: (i) Mild early-dark-energy-like effect, (ii) Late-time spectral correction, (iii) Smoother approach to de Sitter.

11.3 Growth of Structure $f\sigma_8(z)$

Predictions: (i) Scale-dependent growth suppression from Ξ-kernel nonlocal damping, (ii) Mild deviation from ΛCDM at $z < 1$. Testable with RSD, weak lensing, galaxy clustering.

11.4 CMB Signatures

Predictions: (i) Modified ISW signal from smoother $w_{\text{eff}}(z)$ transition, (ii) Sector-dependent correlation anomalies.

11.5 Spectral Signatures (Rivero–Zeta)

Predictions: (i) Time-dependent spectral flow $\frac{d}{dt}\zeta_R(s,t) \neq 0$ during transition, (ii) Late-time spectral freeze as attractor reached, (iii) Sector-resolved spectral fingerprint. (Assumption; quantitative predictions deferred to future work.)

11.6 Nonlocal Correlation Signatures

Ξ-kernel produces: (i) Long-range correlation distortions in lensing/galaxy correlations/CMB, (ii) Scale-dependent deviations in $P(k)$, (iii) Sector-dependent anisotropies.

11.7 Summary of Testable Predictions

Non-monotonic $w_{\text{eff}}(z)$, smoother transition, modified Hubble, scale-dependent growth suppression, ISW anomalies, spectral flow signatures, sector correlation patterns. Quantitative, falsifiable, distinct from ΛCDM.

12. Mathematical Consistency and Theorem Structure

12.1 Preliminaries and Functional Spaces

$(\mathcal{M}, g) = (\mathbb{R}^3 \times \mathbb{R}, -dt^2 + a(t)^2 d\mathbf{x}^2)$.

Variables: $G(t) \in C^2(\mathbb{R})$, $p(t) \in C^1(\mathbb{R})$, $\Xi(t,t') \in L^2(\mathbb{R}^2)$, $a(t) \in C^2(\mathbb{R})$, $\lambda_k(t) \in C^1(\mathbb{R})$.

I_{geom}: Fréchet differentiable, bounded below, coercive.

Theorem 12.1 (Well-posedness). *System*

$$\begin{aligned} &\ddot{G} + 3H\dot{G} + V'(G) = F_\Xi[G] + F_I[G,p] \\ &\dot{p} = -\gamma(p - p_\infty) - \frac{\delta I_{\text{geom}}}{\delta p} \\ &H^2 = \rho_{\text{tot}} \end{aligned} \tag{101}$$

admits unique local solution for admissible initial data. (Picard–Lindelöf)

Theorem 12.2 (Energy Boundedness). *If $\lambda_G > 0$ and Ξ positive semidefinite, then:*

$$E_{\text{tot}} = E_G + E_\Xi + E_p + E_{\text{spec}} \quad \textit{is bounded below} \tag{102}$$

Theorem 12.3 (Existence of Fixed Points). *There exists at least one fixed point $(G^*, p_\infty, H_\infty, \Xi^*, \lambda_{k,*})$ satisfying the master equation.*

Theorem 12.4 (Stability of Attractor). *If:*

$$M_{\text{eff}} = m_G^2 + 2\alpha_{\text{DM}} + 3\lambda_G (G^*)^2 + \left.\frac{\partial^2 I_{\text{geom}}}{\partial G^2}\right|_{G^*} + \lambda_{\Xi,\min} > 0 \tag{103}$$

then the fixed point is asymptotically stable.

Theorem 12.5 (Spectral Continuity). *Eigenvalues $\lambda_k(t)$ of $\mathcal{D}(t)$ depend continuously on t. (Kato perturbation theory)*

Theorem 12.6 (Sector Consistency). *The 33-sector decomposition is preserved: $\Xi_{\sigma\sigma'}(t_0, t_0') = 0$ for $\sigma \neq \sigma'$ implies $\Xi_{\sigma\sigma'}(t,t') = 0$ for all t. The 33-web is dynamically stable.*

13. Synthesis: The AI33-MPOPT Unified Architecture

13.1 The Six-Layer Structure

1. **Local Cosmological Dynamics:** FLRW, Hubble, matter dilution.
2. **Vacuum-Tension Geometry (Garkness):** scalar vacuum, dark-matter, pause-factor.
3. **Nonlocal Correlation Geometry (Ξ):** long-range correlations, geometric constraints.
4. **Geometric Information Functional:** global constraints, constraint selection.
5. **Spectral Geometry (Rivero–Zeta):** spectral operator, eigenvalues, determinant.
6. **Sector Structure (33-Web):** decomposition, sector dynamics, sector spectra.

13.2 Architecture as Directed Graph

Local $\to$ Vacuum-Tension $\to$ Nonlocal $\to$ Information $\leftarrow$ Spectral, all organized by 33-Sector Structure.

13.3 Master Equation as Central Node

$$\mathcal{D}_{\text{local}} + \mathcal{D}_{\Xi} + \mathcal{D}_{\text{spec}} + \frac{\delta I_{\text{geom}}}{\delta G} = 0 \tag{104}$$

receives input from all layers, outputs universe evolution, determines attractor, enforces consistency.

13.4 Attractor as Architectural Endpoint

$$(G, \dot{G}, p) \to (G^*, 0, p_\infty), \quad H \to H_\infty, \quad \lambda_k \to \lambda_{k,*}, \quad \Xi \to \Xi^* \tag{105}$$

Information-selected, spectrally consistent, sector-resolved, stable, globally constrained vacuum state.

13.5 Architecture as Hierarchical Stack

Layer 1: Metric. Layer 2: Vacuum-Tension. Layer 3: Nonlocal. Layer 4: Information. Layer 5: Spectral. Layer 6: Sector.

13.6 Architecture as Research Program

Paper I: Reduced FLRW. Paper II: Rivero–Zeta. Paper III: Information-Modulated Cosmology. Paper IV: Nonlocal Geometry. Paper V: Sector Structure. Paper VI: Master Equation. Paper VII: Observational Signatures.

14. Future Extensions and Open Directions

14.1 Extension A – Full Nonlocal Dynamics of Ξ-Kernel

Dynamical theory:

$$\partial_t \Xi(t, t') = F[G, \Xi, p; \sigma] \tag{106}$$

with symmetry, positivity, sector-block, spectral preservation. (Assumption; proof deferred to future work.)

14.2 Extension B – Covariant Generalization Beyond FLRW

Generalize to: Bianchi I/IX, perturbed FLRW, full 3+1 ADM.

14.3 Extension C – Perturbation Theory and Structure Formation

Full perturbation theory: scalar, vector, tensor perturbations with nonlocal Ξ-correlations, information-modulated vacuum tension, sector-dependent dynamics.

14.4 Extension D – Spectral Flow and Renormalization

Spectral renormalization group:

$$\frac{d}{d \ln k} \lambda_k(t) = R[G, \Xi, p; \sigma] \tag{107}$$

14.5 Extension E – Sector Dynamics and Phase Transitions

Sector transitions, bifurcations, mergers governed by $\Delta\sigma = \mathcal{G}[G, \Xi, p; \sigma]$.

14.6 Extension F – Information-Theoretic Interpretation

Extend I_{geom} to include: entropy-like terms, Fisher information, KL divergences.

14.7 Extension G – Numerical Simulation Program

Simulate: master equation, spectral flow, Ξ-kernel evolution, sector dynamics.

14.8 Extension H – Observational Confrontation

Fit $w_{\text{eff}}(z)$ to supernovae, compute BAO distances, predict ISW, compute $f\sigma_8(z)$.

15. Conclusion of the Master File

15.1 Summary of Architectural Layers

Six interacting layers: Local Dynamics (FLRW backbone), Vacuum-Tension Geometry (Garkness field), Nonlocal Correlation Geometry (Ξ-kernel), Geometric Information Functional (global constraints), Spectral Geometry (Rivero–Zeta), Sector Structure (33-web).

15.2 The Unified Master Equation

$$\mathcal{D}_{\text{local}} + \mathcal{D}_{\Xi} + \mathcal{D}_{\text{spec}} + \frac{\delta I_{\text{geom}}}{\delta G} = 0 \tag{108}$$

with:

$$\frac{\delta I_{\text{geom}}}{\delta p} = 0, \qquad \frac{\delta I_{\text{geom}}}{\delta \Xi} = 0, \qquad \frac{d}{dt}\zeta_R(0,t) = 0 \tag{109}$$

15.3 The Information-Selected Attractor

$$(G, \dot{G}, p) \to (G^*, 0, p_\infty), \quad H \to H_\infty, \quad \lambda_k \to \lambda_{k,*}, \quad \Xi \to \Xi^* \tag{110}$$

Dynamically stable, spectrally consistent, sector-resolved, globally constrained.

15.4 Observational Consequences

Non-monotonic $w_{\text{eff}}(z)$, smoother transition, modified Hubble, scale-dependent growth suppression, ISW anomalies, sector correlation patterns, spectral flow signatures.

15.5 Mathematical Consistency

Well-posedness, energy boundedness, fixed point existence, attractor stability, spectral continuity, sector preservation.

15.6 Conceptual Unity

Synthesis of: geometry (metric, vacuum tension, nonlocal structure), information (global constraints, modulation), spectral theory (Rivero–Zeta), dynamics (master equation, attractors), structure (33-web).

15.7 Closing Statement

AI33-MPOPT provides: new geometric foundation for cosmology, mathematically controlled ΛCDM extension, unified local/nonlocal/spectral/informational treatment, structured extensible architecture. This master file completes the foundational construction of the theory.

A. Notation and Definitions

A.1 Spacetime and Geometry

$(\mathcal{M}, g)$: 4D Lorentzian manifold, signature $(-,+,+,+)$.
FLRW: $ds^2 = -dt^2 + a(t)^2\, d\mathbf{x}^2$.
Hubble: $H(t) = \dot{a}(t)/a(t)$.
Laplace–Beltrami: $\Delta_{g(t)}$.

A.2 Fields and Variables

$G : \mathcal{M} \to \mathbb{R}$ (Garkness vacuum-tension scalar), $p : \mathcal{M} \to \mathbb{R}$ (pause-factor), $\Xi : \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ (nonlocal kernel), $\sigma \in \{1, \ldots, 33\}$ (sector index), $\lambda_k(t) \in \mathbb{R}_+$ (eigenvalues).

A.3 Potentials and Energies

$$V(G) = \Lambda_G + \frac{1}{2} m_G^2 G^2 + \frac{\lambda_G}{4} G^4 \tag{111}$$

$$\rho_{\mathrm{DM}} = \alpha_{\mathrm{DM}} G^2 \quad \text{(emergent)} \tag{112}$$

$$\rho_{\Lambda,\mathrm{eff}} = p(t)\Lambda_{\mathrm{eff}} \tag{113}$$

$$\rho_\Xi = \frac{1}{2a^3} \int G\, \Xi\, G\, dt' \tag{114}$$

A.4 Operators and Functionals

$$(K_\Xi f)(x) = \int_{\mathcal{M}} \Xi(x,y)\, f(y)\, d^4y \tag{115}$$

$$\mathcal{D}(t) = -\Delta_g + M^2 + U_G + U_\Xi \tag{116}$$

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s} \tag{117}$$

$$I_{\mathrm{geom}} = I_G + I_\Xi + I_p + I_\sigma \tag{118}$$

A.5 Sector Decomposition

$$\mathcal{M} = \bigcup_{\sigma=1}^{33} U_\sigma \tag{119}$$

$$\chi_\sigma(x) = \begin{cases} 1 & x \in U_\sigma \\ 0 & \text{otherwise} \end{cases} \tag{120}$$

$$G_\sigma = \chi_\sigma G \tag{121}$$

$$\Xi_{\sigma\sigma'} = \chi_\sigma\, \Xi\, \chi_{\sigma'} \tag{122}$$

A.6 Dynamical Equations

Garkness:

$$\ddot{G} + 3H\dot{G} + (m_G^2 + 2\alpha_{\mathrm{DM}})G + \lambda_G G^3 + \int \Xi\, G\, dt' = -\frac{\delta I_{\mathrm{geom}}}{\delta G} \tag{123}$$

Pause:

$$\dot{p} = -\gamma(p - p_\infty) - \frac{\delta I_{\mathrm{geom}}}{\delta p} \tag{124}$$

Spectral:

$$\frac{d}{dt}\zeta_R(0,t) = 0 \tag{125}$$

A.7 Cosmological Quantities

$$\rho_{\rm tot} = \rho_{m0}\, a^{-3} + \rho_G + \rho_{\rm DM} + \rho_{\Lambda,\rm eff} + \rho_\Xi \tag{126}$$

$$p_{\rm tot} = p_G - p(t)\Lambda_{\rm eff} \tag{127}$$

$$w_{\rm eff} = p_{\rm tot}/\rho_{\rm tot} \tag{128}$$

$$q = \frac{1}{2}(1 + 3w_{\rm eff}) \tag{129}$$

B. Derivations of Key Equations

B.1 Garkness Field Equation

From $S_{\text{local}} = \int d^4x\,\sqrt{-g}\,\left[\frac{1}{2}\dot{G}^2 - V(G)\right]$.

Variation of kinetic term:

$$\delta S_{\text{kin}} = -\int a^3\left[\ddot{G} + 3H\dot{G}\right]\delta G\,dt \tag{130}$$

Variation of potential:

$$\delta S_{\text{pot}} = -\int a^3\,V'(G)\,\delta G\,dt \tag{131}$$

Setting $\delta S = 0$:

$$\ddot{G} + 3H\dot{G} + m_G^2 G + \lambda_G G^3 = 0 \tag{132}$$

B.2 Nonlocal Ξ Contribution

From $S_\Xi = \frac{1}{2}\iint G\,\Xi\,G\,d^4x\,d^4y$.

Variation:

$$\frac{\delta S_\Xi}{\delta G(x)} = \int_{\mathcal{M}} \Xi(x,y)\,G(y)\,d^4y = \mathcal{D}_\Xi[G] \tag{133}$$

B.3 Pause-Factor Equation

Relaxation term:

$$\delta S_{\text{relax}} = -\int a^3\,\gamma(p - p_\infty)\,\delta p\,dt \tag{134}$$

Information contribution:

$$\frac{\delta I_p}{\delta p} = \beta_1 G^2 + \beta_2 \int \Xi\,G \tag{135}$$

Result:

$$\dot{p} = -\gamma(p - p_\infty) - \frac{\delta I_{\text{geom}}}{\delta p} \tag{136}$$

B.4 Friedmann Equation

$$\rho_{\text{tot}} = \rho_{m0}\,a^{-3} + \rho_G + \rho_{\text{DM}} + \rho_{\Lambda,\text{eff}} + \rho_\Xi \tag{137}$$

where $\rho_G = \frac{1}{2}\dot{G}^2 + V(G)$.

Friedmann:

$$H^2 = \rho_{\text{tot}} \tag{138}$$

B.5 Spectral Flow

$$\frac{d}{dt}\zeta_R(s,t) = -s\sum_k \lambda_k^{-s-1}\,\dot{\lambda}_k \tag{139}$$

where:

$$\dot{\lambda}_k = \langle \phi_k, \dot{\mathcal{D}}\,\phi_k\rangle = -2H\,\frac{k^2}{a^2} + \dot{U}_G + \dot{U}_\Xi \tag{140}$$

C. Spectral Operator Details

C.1 Definition

$$\mathcal{D}(t) = -\Delta_{g(t)} + M^2 + U_G(t) + U_\Xi(t) \tag{141}$$

where:

$$U_G = m_G^2 G^2 + \lambda_G G^4 \tag{142}$$

$$U_\Xi = \int \Xi(t,t')\, G(t')\, dt' \tag{143}$$

C.2 Self-Adjointness

Domain: $\mathcal{D}(\mathcal{D}(t)) = H^2(\mathbb{R}^3)$.
Positivity: $\langle f, \mathcal{D}(t) f \rangle \geq M^2 \|f\|^2$.

C.3 Eigenvalues

$\mathcal{D}(t)\,\phi_k = \lambda_k(t)\,\phi_k$.
For flat FLRW: $\phi_k = e^{i\mathbf{k}\cdot\mathbf{x}}$:

$$\lambda_k(t) = \frac{k^2}{a(t)^2} + M^2 + U_G(t) + U_\Xi(t) \tag{144}$$

C.4 Rivero–Zeta

$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s}$ converges for $\mathrm{Re}(s) > \frac{3}{2}$.
Analytic continuation with poles at $s = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, \ldots$
Determinant:

$$\det \mathcal{D}(t) = \exp\left[-\zeta_R'(0,t)\right] \tag{145}$$

C.5 Spectral Flow and Heat Kernel

Flow:

$$\frac{d}{dt}\zeta_R = -s \sum_k \lambda_k^{-s-1} \left[-2H\,\frac{k^2}{a^2} + \dot{U}_G + \dot{U}_\Xi\right] \tag{146}$$

Heat kernel:

$$\zeta_R(s,t) = \frac{1}{\Gamma(s)} \int_0^\infty \tau^{s-1}\, \mathrm{Tr}\left(e^{-\tau \mathcal{D}}\right) d\tau \tag{147}$$

D. Sector Decomposition Proofs

Theorem D.1 (Orthogonality). *For $\sigma \neq \sigma'$:* $\mathrm{supp}(G_\sigma) \cap \mathrm{supp}(G_{\sigma'}) = \varnothing$.

Proof. $\chi_\sigma \chi_{\sigma'} = 0 \Rightarrow G_\sigma G_{\sigma'} = 0$. □

Theorem D.2 (Block Structure). $\Xi = \sum_{\sigma,\sigma'} \Xi_{\sigma\sigma'}$ *with* $\Xi_{\sigma\sigma'} = 0$ *unless* $x \in U_\sigma$, $y \in U_{\sigma'}$.

Theorem D.3 (Preservation). $\Xi_{\sigma\sigma'}(t_0, t_0') = 0$ *for* $\sigma \neq \sigma'$ *implies* $\Xi_{\sigma\sigma'}(t, t') = 0$ *for all* t.

Proof. $\frac{\delta I_{\mathrm{geom}}}{\delta \Xi_{\sigma\sigma'}} = 0$ identically, so $\partial_t \Xi_{\sigma\sigma'} = 0$. □

Theorem D.4 (Sector Decomposition of Dynamics). *Sector dynamics decomposes into 33 independent equations.*

Theorem D.5 (Spectral Decomposition).

$$\mathcal{D}(t) = \bigoplus_{\sigma=1}^{33} \mathcal{D}_\sigma(t) \tag{148}$$

Appendix G — Extended Notes on Section 13.7

This continuation is provided solely to prevent misclassification of Section 13.7 and to anchor its language within established Kaluza–Klein terminology familiar to gr-qc readers. Section 13.7 in the main manuscript provides only a brief statement of the effective higher-dimensional interpretation; the detailed clarification is provided here.

(Companion file to "AI33-MPOPT: Garkness Geometry and Information-Modulated Cosmology" — not a standalone document)

This file continues Section 13.7 of the main AI33-MPOPT manuscript. It introduces no new physics, no new equations, and no new assumptions. Its only purpose is to clarify the Kaluza–Klein-type structural interpretation referenced in Section 13.7.

13.7 — Effective Higher-Dimensional Interpretation (Extended Notes)

The parallels described below arise because the mathematical structures inside AI33-MPOPT match those produced by Kaluza–Klein-type reductions, even though no extra dimensions are assumed.

13.7.A — Why a Kaluza–Klein Interpretation Applies

Kaluza–Klein (KK) reductions generate four characteristic structures:

1. A radion/modulus scalar
2. A bulk-induced nonlocal kernel
3. A discrete spectral tower
4. A sector-based mode decomposition

AI33-MPOPT contains all four, but they arise internally within the 4D theory. This is why Section 13.7 mentions KK-type behavior.

13.7.B — Garkness $\leftrightarrow$ Radion-Like Degree of Freedom

In KK reductions, integrating out hidden geometry produces a scalar field that modulates geometric tension. AI33-MPOPT already contains such a scalar:

$$G(x)$$

It:

- modulates vacuum tension,
- contributes dynamically to the energy budget,
- couples to nonlocal and spectral layers.

This is exactly the functional role of a radion in KK theory—but here it arises within 4D, not from an extra dimension.

13.7.C — E-Kernel ↔ Bulk-Induced Nonlocal Operator

KK and braneworld reductions generate bilocal terms of the form:

$$\int d^4x\, d^4y\, G(x)\, K(x,y)\, G(y)\,.$$

AI33-MPOPT contains:

$$E(x,y)\,,$$

a symmetric, positive, bilocal, non-signaling kernel. Its mathematical structure matches that of a bulk-induced Green's function, even though AI33-MPOPT is purely four-dimensional.

13.7.D — Rivero–Zeta ↔ Generalized KK Spectral Tower

KK theories produce discrete mass towers:

$$m_n^2 = m_0^2 + \frac{n^2}{R^2}\,.$$

AI33-MPOPT contains a spectral operator:

$$D(t)\,,$$

with eigenvalues $\lambda_k(t)$ summarized by the Rivero–Zeta function:

$$\zeta_R(s,t) = \sum_k \lambda_k(t)^{-s}\,.$$

This acts as a generalized KK tower, producing an effective spectral dimension > 4 without introducing any new coordinates.

13.7.E — 33-Web ↔ Mode-Space Decomposition

KK towers naturally split into symmetry sectors or mode families. AI33-MPOPT already contains:

$$33 \text{ sectors}\,,$$

arising from the block structure of the E-kernel. This is mathematically identical to mode-space partitioning in KK reductions.

13.7.F — Master Equation ↔ KK-Reduced Effective Form

The AI33-MPOPT Master Equation combines:

- local FLRW dynamics,
- nonlocal E-kernel contributions,
- a spectral tower,
- sector structure,
- and global information constraints.

This is the same combination that appears in effective KK-reduced gravity:

- 4D Einstein equations,
- bulk-induced nonlocality,

- KK mass tower,
- mode-sector structure,
- radion/modulus constraint.

AI33-MPOPT reproduces this structure internally, without invoking extra dimensions.

13.7.G — Scope Clarification

This continuation:

- does not introduce extra dimensions,
- does not assume compactification,
- does not add new physics,
- does not modify the Master Equation.

It only clarifies why Section 13.7 states that AI33-MPOPT "admits an effective higher-dimensional interpretation."

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